

Duke Tran

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EDUCATION

College of William & Mary

Bachelor of Science in Computer Science, Minor in Finance

Williamsburg, VA

Graduated Jan 2023

- **GPA:** 3.82 / 4.0 (Major: 3.93 / 4.0)
- **Honors:** Summa Cum Laude, Dean's List (Spring '20, Fall '20, Spring '21, Fall '21, Spring '22, Fall '22)

PROFESSIONAL EXPERIENCE

Amazon Web Services (AWS)

New York, NY

Software Development Engineer I (AI/ML)

Oct 2023 – Present

- Scaffolded critical UI for AWS Q Business console prior to GA launch, implementing 30+ UX enhancements. Implemented the Q Metadata Boosting split panel and an internal web app for chat events inspection post-GA.
- Worked on development of the subscriptions feature end-to-end, enabling organizations to organize and subscribe users to Q Business. Contributed to API service code, console UI, and gameday testing.
- Implemented HTML table parsing to enable tabular search and web app UI changes to better render and display tables. Ported a critical HTML parser from Java to Python to conduct accuracy testing.
- Supported a variety of ops work, ranging from fixing the delete subscriptions workflow, CloudTrail LaserHouse onboarding, stabilizing pipeline security tests, to API FAS Credentials encryption. Downgraded pre-prod tickets to lower severity and observed a 90+% reduction in weekly oncall ticket load.
- Spearheaded efforts to make owned pipelines fully CI/CD, including fixing existing pipeline integration tests, fixing CloudWatch alarm definitions and noisiness, and enabling auto-rollback on aggregate monitors.

Arch

New York, NY

Full Stack Software Engineer

May 2023 – Sep 2023

- Maintained and optimized code relating to complex ownership of investments, an integral part of the platform offering. Wrote unit tests for a particular code path achieving 90+% coverage.
- Spearheaded projects to implement features for the Addepar integration used by at least 25% of existing clients. Optimized a critical SQL query to halve average runtime, down to around 2.3 seconds from 4.
- Designed and implemented a full project to enable exporting of financial commitments to Addepar, a feature specifically requested by a number of core clients.
- Engineered a historical financials selector component, from the backend to the UI, that enabled clients to view their investments' financials through a historical period of time, used daily by hundreds of users.

NSF REU Site: HUMANS MOVE

Fairfax, VA

Research and Software Development Intern

Jun 2021 – Aug 2021

- Developed evasive algorithms paired with the SHARKS (Secure, Heterogeneous, Autonomous, and Rotational Knowledge for Swarms) Protocol to defend a UAV swarm against adversarial UAV attacks.
- Simulated UAV swarms in VR and Unity 3D to gather data on effectiveness of evasive maneuvers.
- The developed algorithm yielded an error of 1.58%, a 79% improvement compared to the baseline swarm with no ejection algorithm, for mean average distance which was the measure of evasion effectiveness.

Whisperings

Fairfax, VA

Mobile App Development Intern

Jul 2020 – Aug 2020

- Learned to work with Swift, XCode, and SQLite for iOS development.
- Developed a fully-fledged iOS app with playback and user-recording capabilities.
- Created and maintained a loading and databasing system to store sounds (whispers).

SKILLS, ACTIVITIES & INTERESTS

Technical Skills: Python, Java, Javascript, HTML, CSS, MySQL, PostgreSQL, MongoDB, NodeJS, React, Typescript, C/C++, Go, Docker, RabbitMQ, GraphQL, Ruby, AWS

Interests: Software Development/Engineering, Finance and Financial Analysis, Quantitative Finance, AI/ML